MANA VOICE AI

Real-Time Multilingual Voice Communication

Project Presentation | Vijjapu Nandini

Good morning everyone.

My name is Nandini, and I am currently pursuing my B.Tech in Computer Science and Engineering with Artificial Intelligence. As part of my internship, I am developing a project called **ManaVoice AI**.

ManaVoice AI is a real-time multilingual voice communication platform that enables two people speaking different languages to communicate naturally.

**Problem Statement

** Imagine two people having a conversation. One speaks Telugu, while the other understands only English. They face difficulties communicating because of language differences. Existing translation applications often require users to manually translate messages, which interrupts the natural flow of conversation.

I wanted to solve this problem by creating a platform where two people can speak in their own languages and hear each other's messages in their preferred language.

How My Project Works

ManaVoice AI provides a shared conversation room for two participants. Both users join the same room, select their preferred languages, and start communicating through their microphones.

For speech recognition, I integrated **Gnani Prisma**, which converts the participant's spoken audio into text. After that, the recognized text goes through the translation process, and the translated text is passed to **Timbre TTS** to generate voice output.

For translation, I implemented the **NLLB-200 Distilled 600M model**, which performs local translation between the supported languages.

The generated WAV audio is then delivered to the other participant and played automatically. The same process works in the opposite direction, enabling two-way communication.

Technologies Used

For the frontend, I used React.js and JavaScript to build the user interface and handle microphone input and audio playback.

For the backend, I used Python and FastAPI to manage application processing and integrate the AI services.

I implemented WebSockets for live communication and audio transmission between participants. Gnani Prisma handles speech-to-text, NLLB-200 handles translation, and Timbre TTS generates the translated voice.

One important implementation detail is that microphone audio is transmitted continuously in small frames, while translation and voice generation take place for each completed speech segment.

Conclusion

Building ManaVoice AI gave me practical experience in integrating AI models and APIs into a working full-stack application. I learned how different technologies can work together to solve a real-world communication problem.

In the future, I aim to support more languages, improve translation speed, and make voice communication even more seamless.

My vision is simple:

Your Language Should Never Limit Your Potential.

Thank you!